? SQL

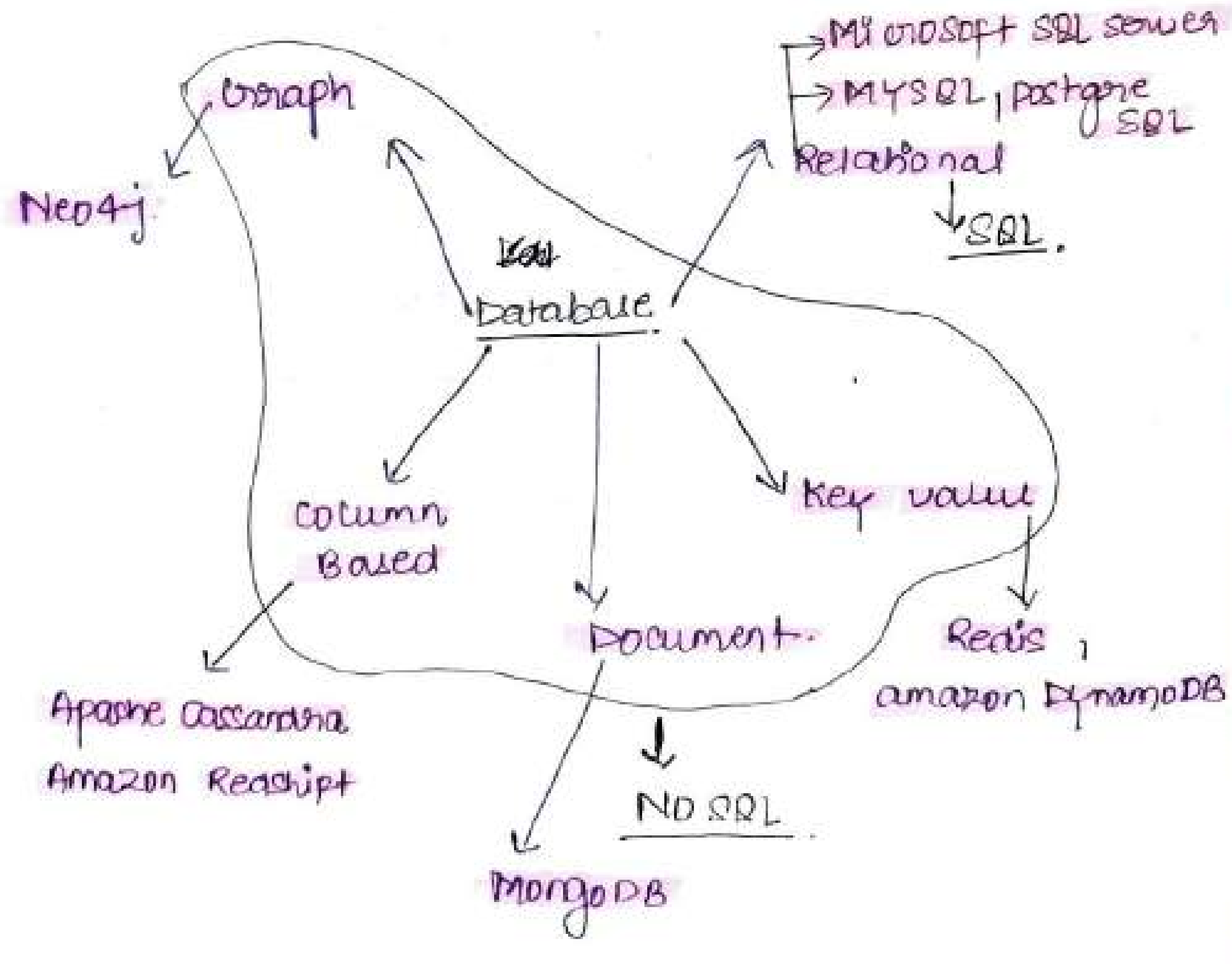
Data with Baraa

Begineer Level

Structured Query Language

- Secure
- Big Data handle
- Easy to Manipulate

- Database stores Data
- SQL speak to Database
- DBMS Manages Database
- Server where database lives.



SERVER

- DATABASE
  - Schema (order)
    - (items) Table
    - Table (purchase)

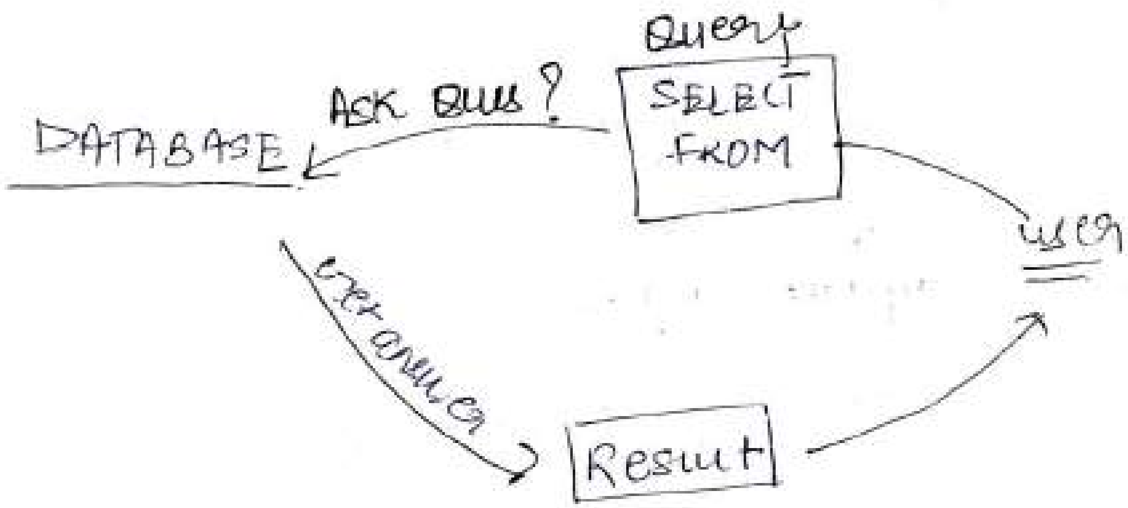
Data types
- Numeric
  - Int
  - decimal
- Text (String)
  - Char (fixed)
  - Varchar
- Date & Time
  - Date
  - Time

TYPES OF SQL Commands.

- Data Defination Language(DDL)
  - Create
  - Alter
  - Drop
- Data Manipulation Language(DML)
  - Insert
  - update
  - Delete.
- Data Query language :- SELECT

SQL SELECT

Ask your Data



SQL Query

SELECT *

- Choose columns

  - SELECT * (All)

    Retrieves All columns (Everything)

  - FROM

    Tells SQL where to find your data.

For comment use.

```
-- comment
/* comment */
```

- SELECT FEW COLUMNS

Pick only the columns you Neded instead to all.

```
② SELECT
    COL1,
    COL2
① FROM Table.
```

Example.

```
SELECT
    name,
    country
FROM customer.
```

→ No comm after last column.

- ~~SELECT~~ SQL QUERY WHERE

- Filter your Data.

Where :- Filter your based on a condition.

(3) SELECT *
(1) FROM Table
(2) WHERE condition.

~~Example~~ OR

(3) SELECT
name,
country
(1) FROM Table
(2) where condition.

Example
- Score not equal to 0

```
SELECT *
  FROM customers
  WHERE score != 0
```

$\underbrace{\text{score != 0}}_{\text{condition.}}$

- customer from Germany

```
SELECT *
  FROM customer
  WHERE country = 'Germany'
```

$\underbrace{\text{country = 'Germany'}}_{\text{condition.}}$

String - should be in
single quotes.

• SQL Query - order By

↓ sort your data.

ASC   DESC

③ SELECT *

① FROM Table

② ORDER BY score DESC

↓ default order is ASC

Example. sort the Result highest score first

```
SELECT *
FROM customers
ORDER BY score DESC
```

DESC ↓ descending.

• Nested (order By)

```
SELECT *
FROM customers
ORDER BY country ASC, score DESC
```

• SQL Query — Group By
-aggregate your data

combines rows with sames values.

Aggregates a column by another column.

Total score by country

Syntax.

③ SELECT
country, /category.
SUM(score) —aggregation.
① FROM Table
② Group By country

Ex:- find total score by each country.

```
SELECT
country,
SUM(score) AS Total-score
FROM customer
Group By country.
```

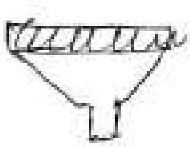
For column name.

Rule :- All columns in the SELECT must be either aggregated or included in GROUP BY.

- SQL Query - HAVING
  - Filter aggregate data.
  - Filter data after aggregation.
  - Can be used only with GROUP BY.

Syntax

④ SELECT
    country,
    SUM(score)
① FROM Table
② GROUP BY country
③ Having condition,
    SUM(score) >800



Filter Your Data

① Before aggregation.

② After aggregation.

⑤ SELECT
country,
SUM(score)

① FROM Table

Before aggregation. ← ② WHERE score > 400

③ GROUP BY country

④ HAVING SUM(score) > 800

Ex. Find avg score for each country considering only customer with score not equal to 0 and return only those countries with avg(score) > 430.

```
SELECT
    Country,
    AVG(score) AS avg_score
FROM customers
WHERE score != 0
GROUP BY country
HAVING AVG(score) > 430.
```

• SQL Query – DISTINCT

↓

Remove duplicates

↓

Repeated values

(2) SELECT DISTINCT

country

(1) FROM Table

Example :- Return unique list of all countries.

```
SELECT DISTINCT
        country
FROM customers.
```

• Top – SQL Query

↓

Restrict the Number of items Returned.

```
(2) SELECT TOP 3 (3)
        *
(1) FROM Table
```

Execution order v/s coding order

(5)

Filter columns ← SELECT DISTINCT TOP 2 → Filter Result Row

(7)

Filter Duplicates

col1,

SUM(col2)

(1) FROM Table

WHERE col = 10 (2)

Filter before aggregation

GROUP BY col1 (3)

(4) HAVING SUM(col2) > 50

Filter after aggregation.

ORDER BY col1 ASC

(6)

Execute order

1. FROM
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT DISTINCT
6. ORDER BY
7. TOP

Data Defination language (DDL)

CREATE

Create a new table called persons with columns: id, person_name, birth_date, and phone.

```
CREATE TABLE persons (
    id NOT NULL,
    person_name VARCHAR(50) NOT NULL,
    birth_date DATE,
    phone VARCHAR(15) NOT NULL,
    CONSTRAINT pk-person PRIMARY KEY (id)
)
```

ALTER

Add new column called Email to the persons table.

```
ALTER TABLE persons
ADD email VARCHAR(50) NOT NULL
```

```
ALTER TABLE persons
DROP COLUMN phone.
```

DROP

Remove Table

Delete the Table persons from database.

DROP TABLE persons;

it will delet all Table with Record.

DML

↓ Data Manipulation Language

INSERT

1) INSERT : Manual Entry (Values)

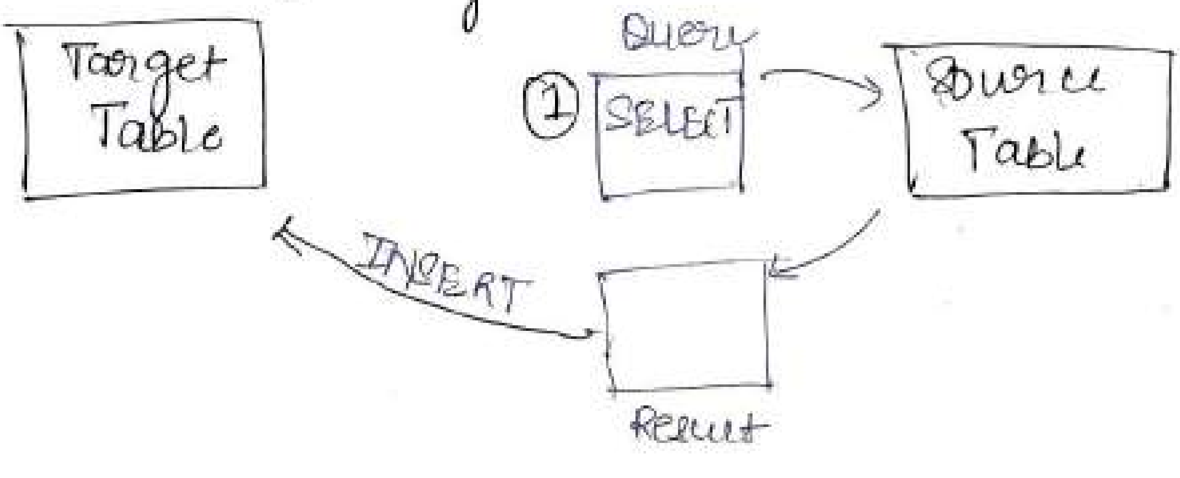
Syntax.

INSERT INTO table_name(col1, col2, col3, ---)
VALUES (value1, value 2, value 3)

Rule: Match the No. of colums and values.
we can do multiple inserts.

- constraints must be fulfilled.

2) using - SELECT



INSERT USING SELECT

Ex:- Copy data from 'customers' table into 'persons'

```
INSERT INTO persons (id, person_name,
                     birth_date, phone)
```
persons → insert data in persons table

```
SELECT
      id,
      first_name,
      NULL,
      'unknown'
FROM customers
```
customers → from customers Table.

UPDATE

Update syntax

~~Upda~~ UPDATE Table_name
SET Col1 = Value1,
Col2 = Value2
WHERE<condition>

[**Note:** Always use where to avoid updating all Rows unintentionally.]

Null :- unknown / No value

Ex:- change the customer score from 6 to .

UPDATE customers
SET Score = 0 → do score 0
WHERE id = 6 where id is 6.

Ex:- Update country and score

UPDATE customers
SET Score = 0
Country = 'UK'
WHERE id = 10

we update two column where id is 10

& Update call

Ex:- update all customer with NULL score by setting their score to 0.

```
UPDATE customers
    SET score = 0        → score 0 kardo
WHERE score IS NULL
```

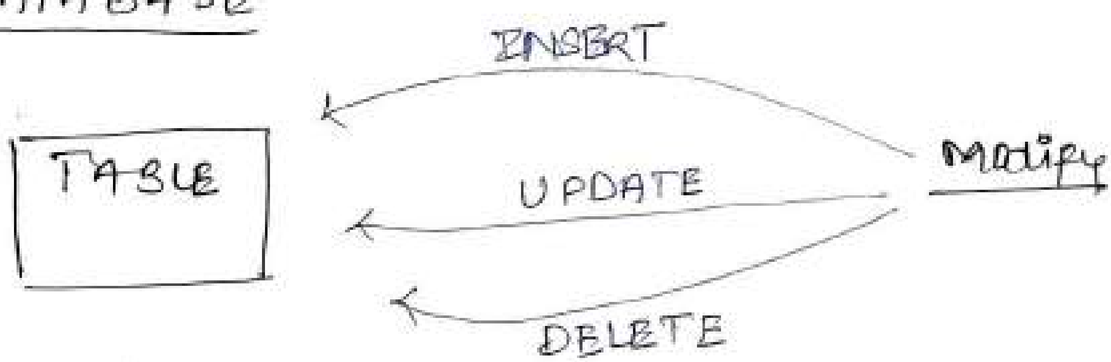
$\underbrace{\text{IS NULL}}_{\text{Jahan Jahan value = NULL hai}}$

Jahan Jahan

value = NULL hai

DELETE

DATABASE



Delete Syntax

```
DELETE FROM table_name
WHERE <CONDITION>
```

Note: Always use WHERE to avoid DELETING all rows unintentionally.

Ex:- Delete all customers with an ID greater than 5.

```
DELETE FROM Customers — Table name
WHERE id > 5
```

$\underbrace{\text{WHERE id > 5}}_{\text{condition}}$

Ex:- Delete all data FROM persons Table.

* $\boxed{\text{TRUNCATE}}$: Clears the whole table at once without checking or logging.

```
TRUNCATE TABLE persons.
```